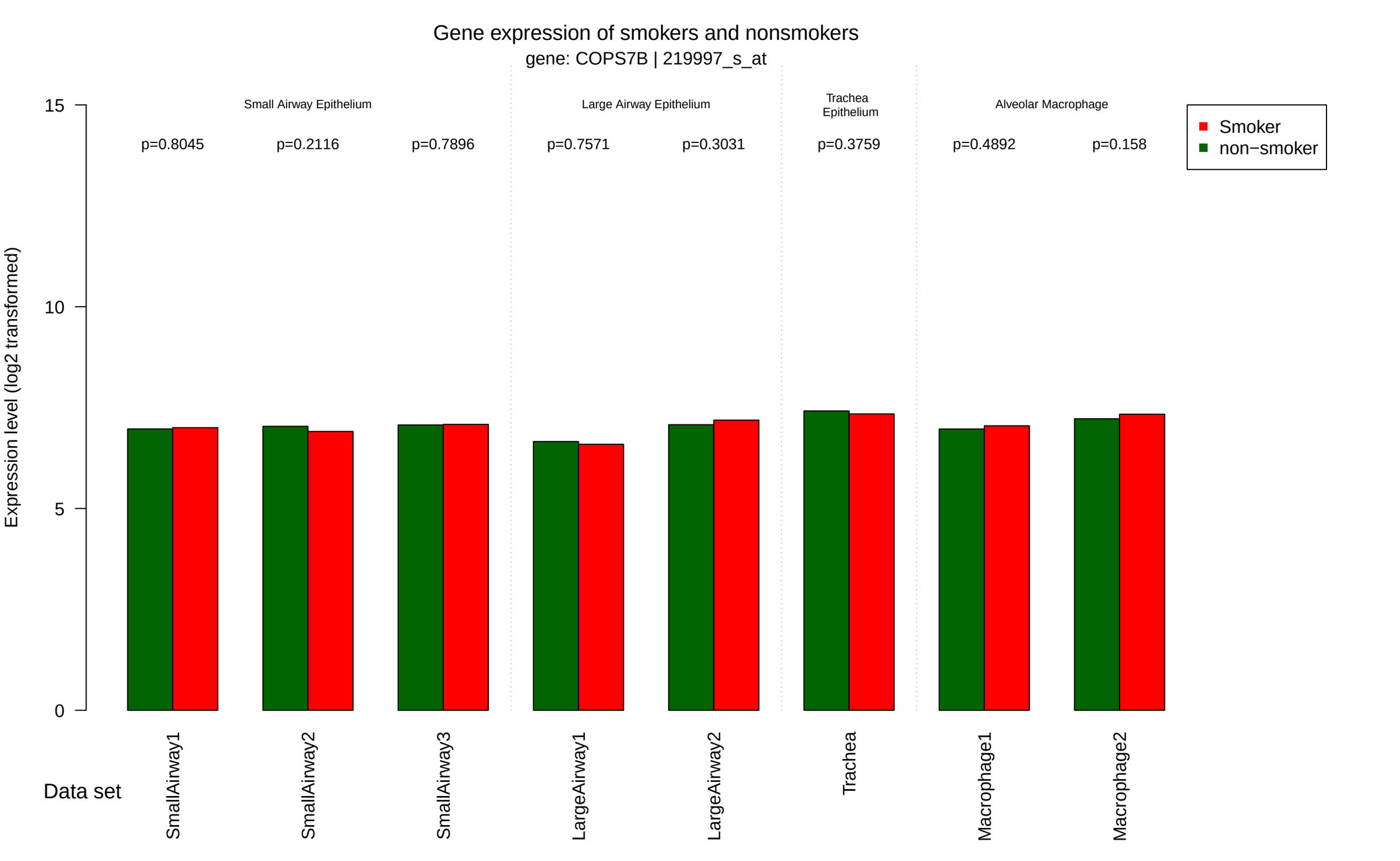
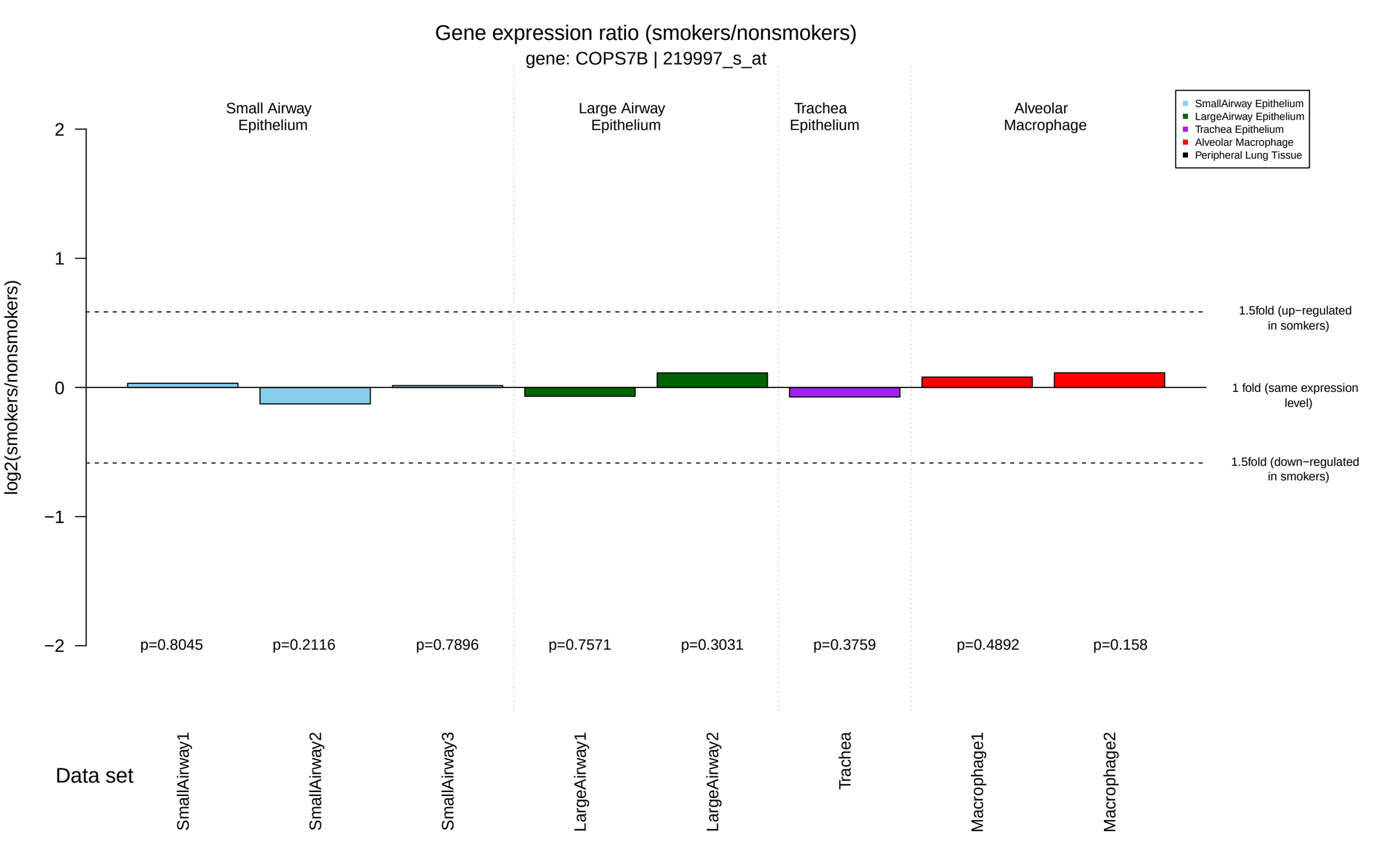
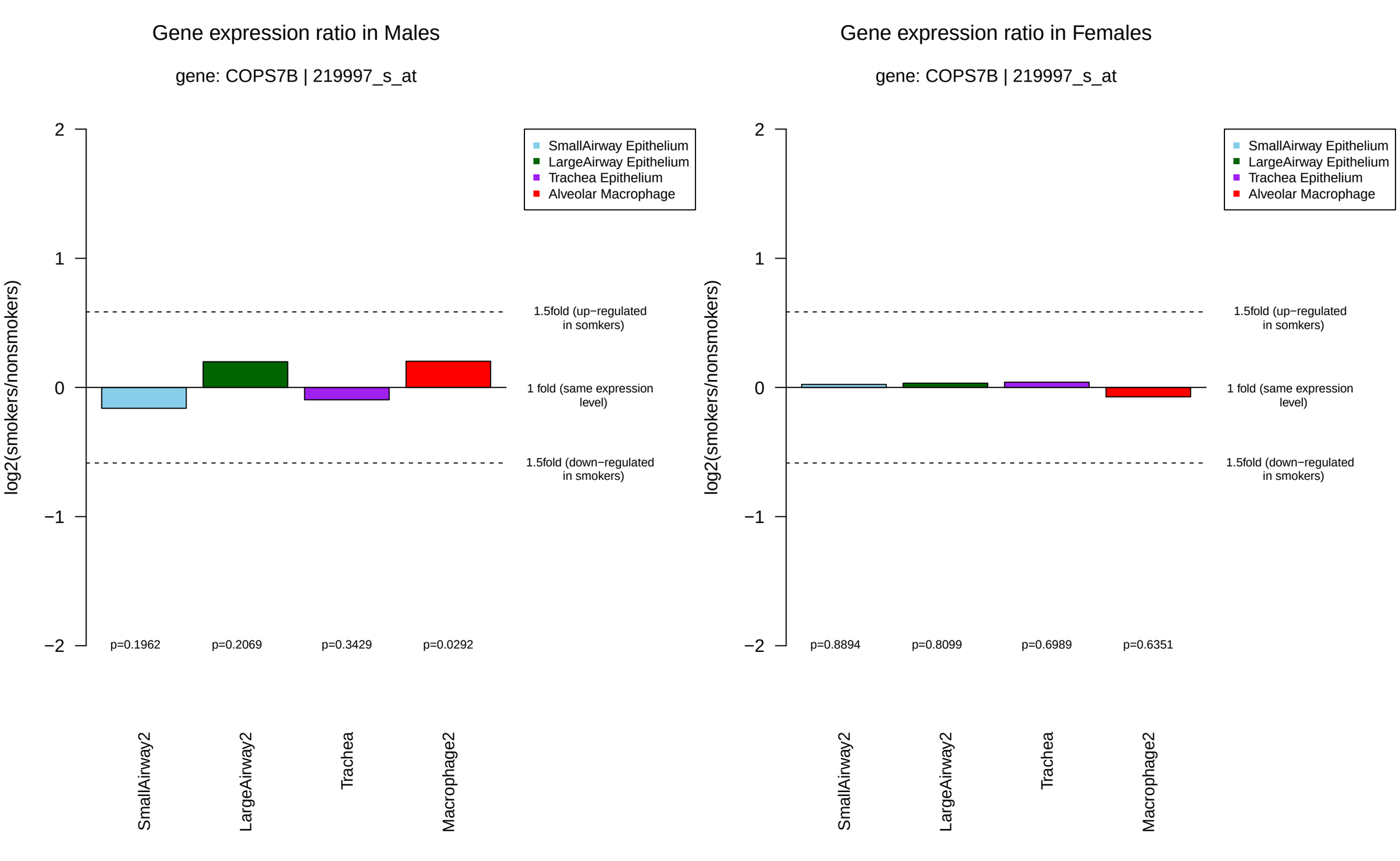
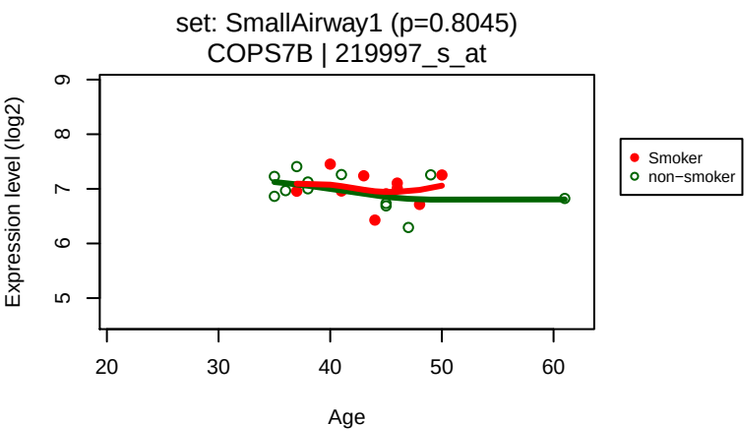
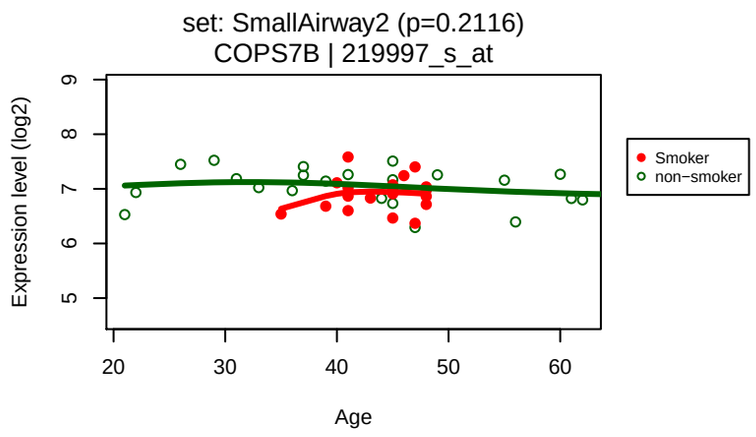
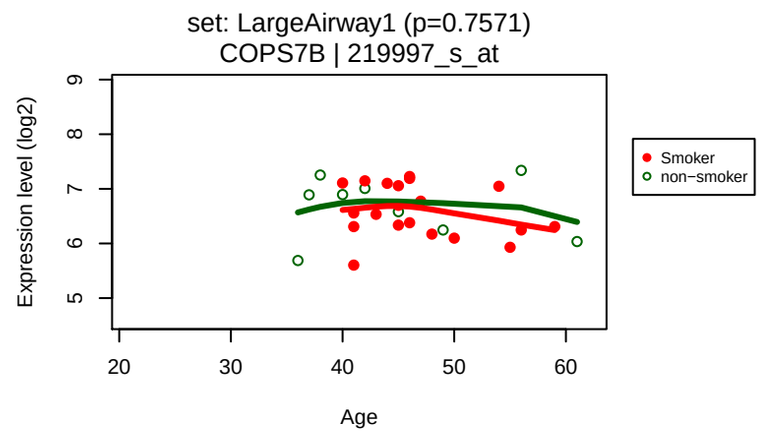

Gene expression of smokers and nonsmokers
gene: COPS7B | 219997_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8045
p=0.2116
p=0.7896
p=0.7571
p=0.3031
p=0.3759
p=0.4892
p=0.158
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

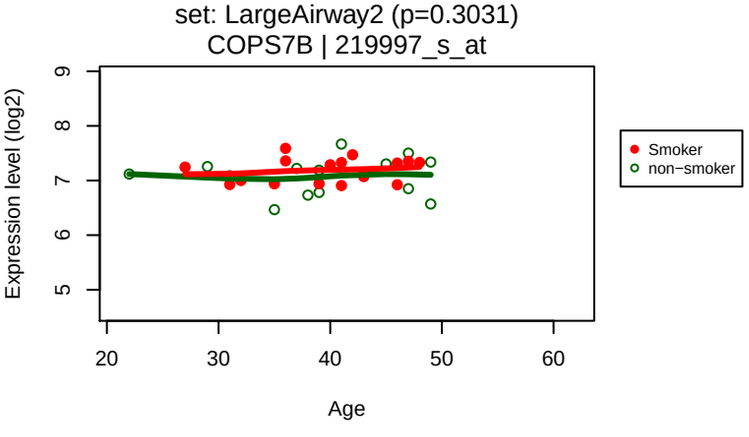

Gene expression ratio (smokers/nonsmokers)
gene: COPS7B | 219997_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8045
p=0.2116
p=0.7896
p=0.7571
p=0.3031
p=0.3759
p=0.4892
p=0.158
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

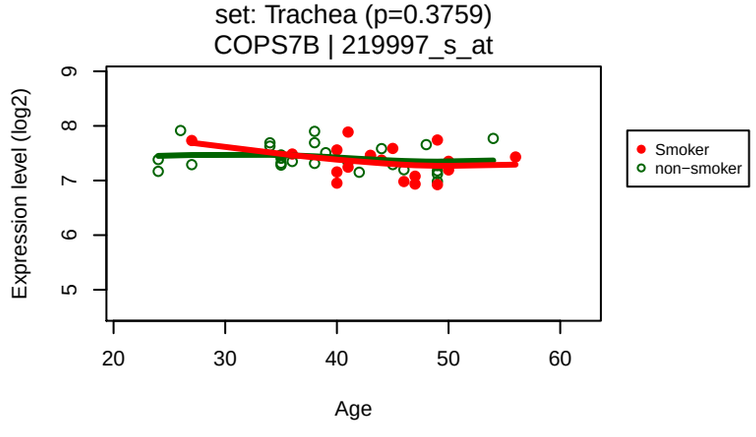

Gene expression ratio in Males
gene: COPS7B | 219997_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1962
p=0.2069
p=0.3429
p=0.0292
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COPS7B | 219997_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8894
p=0.8099
p=0.6989
p=0.6351
SmallAirway2
LargeAirway2
Trachea
Macrophage2

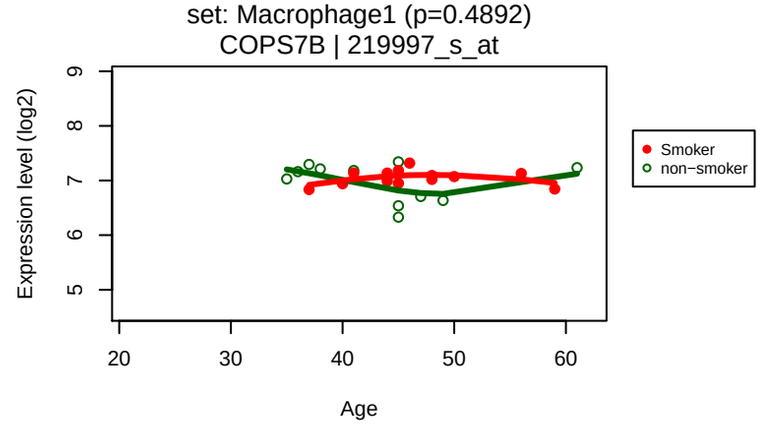

set: SmallAirway1 (p=0.8045)
COPS7B | 219997_s_at
Expression level (log2)
Age
Smoker
non−smoker

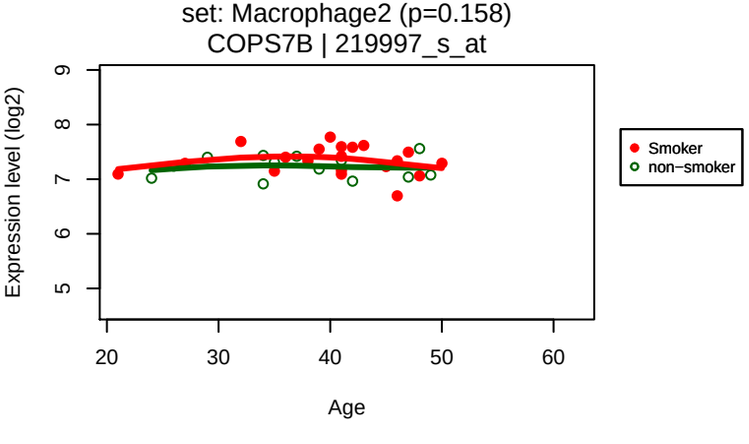

set: SmallAirway2 (p=0.2116)
COPS7B | 219997_s_at
Expression level (log2)
Age
Smoker
non−smoker

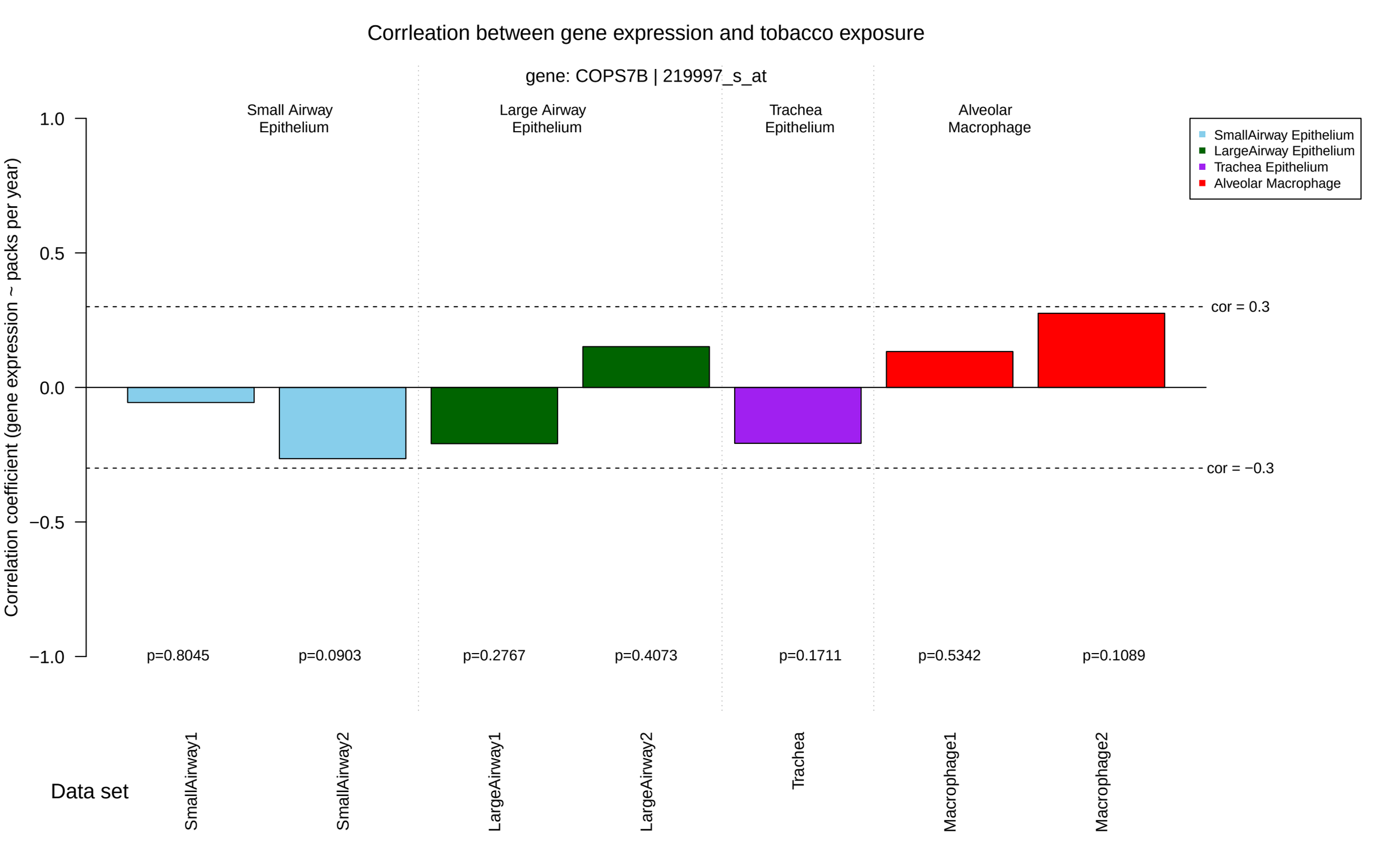

set: LargeAirway1 (p=0.7571)
COPS7B | 219997_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3031)
COPS7B | 219997_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3759)
COPS7B | 219997_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4892)
COPS7B | 219997_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.158)
COPS7B | 219997_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COPS7B | 219997_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8045
p=0.0903
p=0.2767
p=0.4073
p=0.1711
p=0.5342
p=0.1089
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2